

Prathik Pugazhenth

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Experience

RadioFX Inc.

Software Engineering Co-op (AI + DevOps)

Chicago, IL | Jun 2025 – Dec 2025

- Architected and built RadioFX's **GCP/DevOps infrastructure from the ground up**, spanning **2 GKE clusters**, **3 environments**, and **CI/CD for 12 microservices**.
- Automated infrastructure and deployments with **Terraform**, **Argo CD**, **Helm**, and **GitHub Actions**, cutting deployment time **75% (40 → 10 min)**.
- Designed production **cloud networking** across load balancers, firewall rules, Kubernetes NetworkPolicies, and NGINX ingress, sustaining **99.9% availability**.
- Built **Prometheus + Grafana observability** across dev, pre-prod, and prod for service health, latency, and resource monitoring.
- Secured CI/CD across **14 application and infrastructure repositories** with **GCP Workload Identity Federation**, eliminating long-lived service-account credentials.
- Architected an **AI website generation pipeline** converting a client URL into a deployable site in **10–15 min**, orchestrating Gemini audits, v0 generation, RadioFX API integration, and GitHub Actions deployment.
- Engineered a **Kafka-backed API platform on GCP** with rate limits and quota enforcement for **15+ enterprise customers**, sustaining **sub-200ms p95 latency**.
- Engineered a schema-driven **NestJS + Cassandra CMS**, enabling non-technical teams to publish and update content on demand without developer intervention or release-cycle dependency.

Bajaj Finserv Health

SDE II (Full Stack)

Pune, India | Jan 2022 – Jul 2024

- Re-architected **4–5 production microservices** with **Azure Service Bus** event-driven messaging across appointment, consultation, and notification workflows serving **20K+ monthly users**.
- Engineered **specialty-specific EMR** for pediatricians and dentists, extending the physician EMR with clinical workflows for prescriptions, diagnoses, vitals, growth charts, and pediatric immunization schedules.
- Built a **two-way digital consent workflow** for EMR access using deep links and persisted signatures, initially targeting **~16K doctors (20% of the network)**.
- Architected a medical image annotation platform supporting a network of **80,000+ doctors** using **React**, **WebGL**, **NestJS**, and **SQL**.
- Reduced production bugs **~90%** on the consumer platform and **~55%** on the doctor platform over **14 months** through ELK-driven root-cause analysis.
- Re-engineered build and repository architecture across **4 business verticals** using **Turborepo + Git submodules**, cutting build time **25%** and repository/deployment overhead **~60%**.
- Drove a **Next.js SSR + AMP performance overhaul**, improving PageSpeed **65 → 99**, Lighthouse **34 → 88**, and reducing page-load time **85%**.
- Built **SonarQube + ELK** quality and monitoring workflows, increasing code coverage from **0% → 55%** and improving production error visibility.

Projects

NLGraph – Context Graph & Retrieval System

[GitHub](#) | May 2026

Python, spaCy, bge-m3, Qdrant, FAISS, Ollama, SetFit, NetworkX, SQLite

- Constructed a **12-stage NLP-first pipeline** that resolves raw queries through phrase splitting, capability projection, dependency detection, and DAG validation **before any LLM call**.
- Built a **dual-store memory layer** using **Qdrant** for persistent plan caching with **bge-m3** embeddings and **FAISS** for real-time in-session retrieval within a **3,000-token context budget**.

Orchon – Multi-Agent AI Dev Orchestration Platform

[GitHub](#) | Apr 2026

Python, LangChain, PostgreSQL, Redis, GitHub API

- Designed a **DAG-based task decomposition engine** that breaks development requests into role-scoped planner, backend, frontend, and QA subtasks with **isolated per-agent context**.
- Implemented a **human-in-the-loop consensus and merge pipeline** with approval gates and checkpoint resumption, plus provider routing that automatically reroutes on **rate limits or failures**.

Technical Skills

- Languages & Systems:** Python, TypeScript, JavaScript, SQL, Bash, Shell Scripting, Linux, NestJS, Node.js, Fastify, REST APIs, GraphQL
- Cloud Platforms & Infrastructure:** GCP, AWS, Azure, Kubernetes, GKE, Docker, Terraform, Helm, NGINX, Load Balancing, VPC Networking
- CI/CD Pipelines & Platform:** GitHub Actions, Jenkins, Argo CD, GitOps, KEDA, Workload Identity Federation, NetworkPolicies, Secrets Management
- Observability & Messaging:** Prometheus, Grafana, OpenTelemetry, ELK Stack, Kafka, RabbitMQ, Azure Service Bus
- AI & Agentic Systems:** LangChain, RAG, Vector Search, Embeddings, Multi-Agent Orchestration, DAG Task Decomposition, Gemini API, OpenAI API
- Databases:** PostgreSQL, MongoDB, Redis, Cassandra, Qdrant, FAISS, ChromaDB

Education

University of Illinois Chicago

Aug 2024 – May 2026

Master of Science, Computer Science | GPA: 4.00/4.00

Vellore Institute of Technology

Jul 2018 – May 2022

Bachelor of Technology, Computer Science and Engineering | GPA: 3.60/4.00